

Unnamed_Unit

Unit 5 - Knowledge Check

PDF Documents

PDF Document 1: 1738213492160-Chapter 5 - Unit 5 - Knowledge Check.pdf

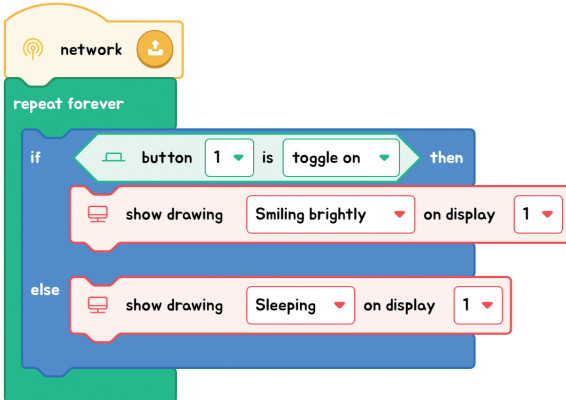
This PDF document is included in the unit materials.



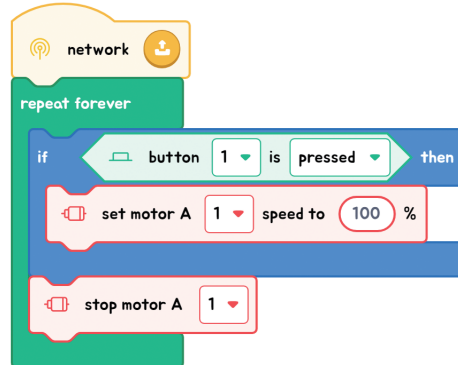
UNIT 5 KNOWLEDGE CHECK

- 1 When moving the motor to the desired position, we also want to control the speed. Let's modify the code using the block below.

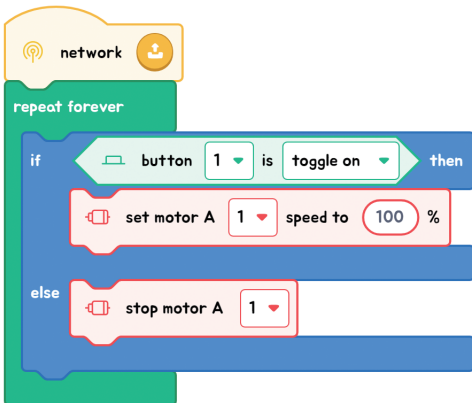
A



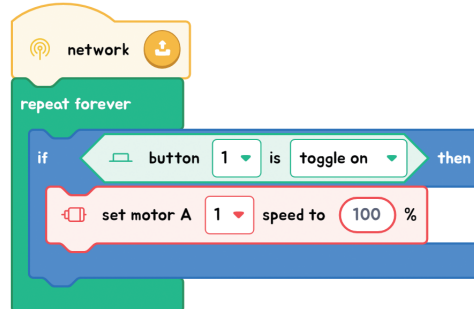
C



B



D



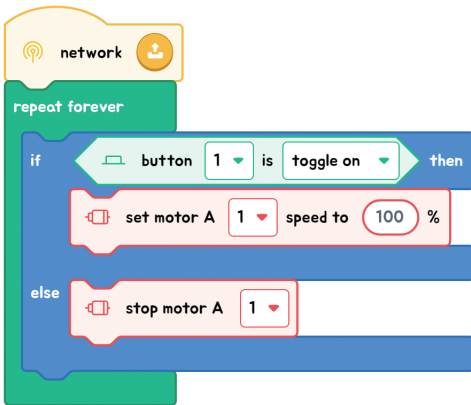
CORRECT ANSWER

D

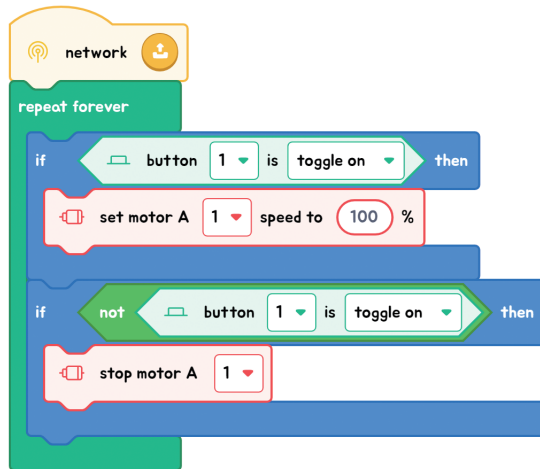


2 What is wrong with the way the toggle-off function is implemented?

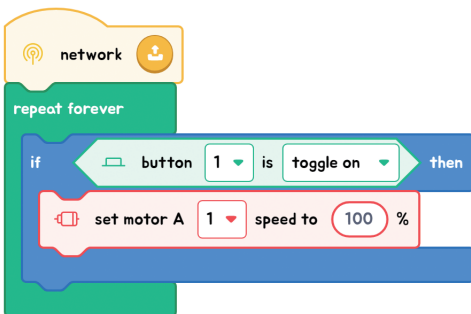
A



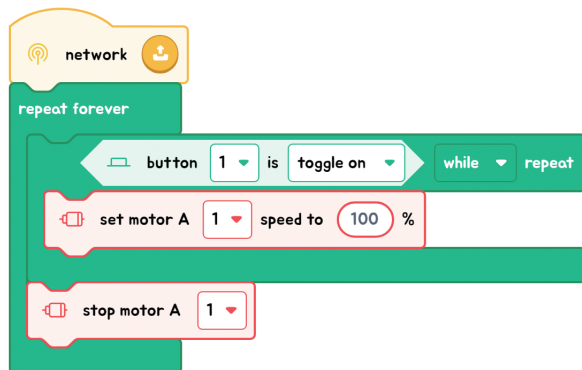
C



B



D



CORRECT ANSWER

C

3 Which of the following is an obtuse angle?

- A** 90°
- B** 45°
- C** 120°
- D** 180°

CORRECT ANSWER

C

4 What is the angle called when two lines meet at 90 degrees?

- ☐ A Acute angle
- ☐ B Right angle
- ☐ C Obtuse angle
- ☐ D Straight angle

CORRECT ANSWER

B

5 Which block would you use to rotate a motor by 45 degrees?

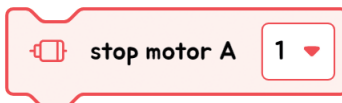
A



B



C



D



CORRECT ANSWER

A

6 What happens if you don't include a "stop" block for a motor in a robot?

- ☐ A The motor will run continuously.
- ☐ B The motor will not start.
- ☐ C The motor will stop immediately.
- ☐ D The motor will reverse direction.

CORRECT ANSWER

A

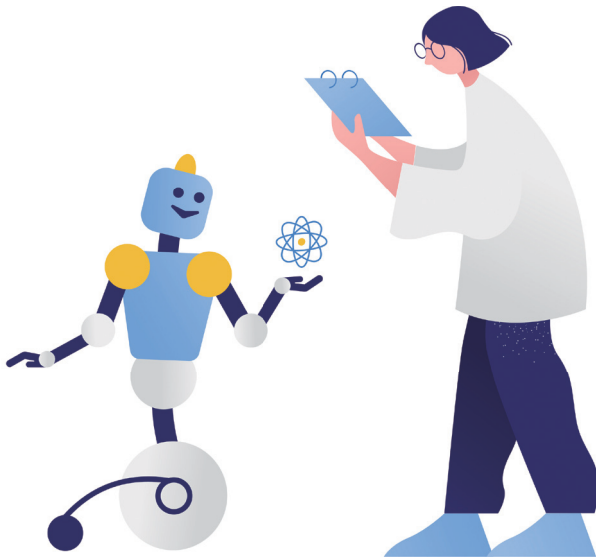


Critical Thinking Challenge

- 1 Imagine that a robot needs to clean an irregularly shaped surface, such as a window with multiple curves. Consider how you can modify the robot's movement strategy or coding approach to ensure that it can efficiently clean all parts of the surface, even when the angles are unpredictable.

- 2 How can I change the angle of the robot to increase cleaning efficiency?

EXPANDING KNOWLEDGE ON THE TOPIC

**1. Why Do Robots Move in Different Directions?**

Have you ever wondered why a robot turns left, sometimes right, and sometimes spins around? Just like fire has different colors, a robot can move in different directions because of the angles we tell it to follow. Let's explore how angles help robots move in a straight line or turn corners and how this can be used in cool projects!

2. How Do Robots Know When to Stop or Turn?

Robots don't have eyes like we do, so how do they know when to stop or make a turn? It's all thanks to sensors and angles! When robots detect a particular object or surface, they use angles to change direction. Let's learn how sensors and angles work together to help a robot make intelligent moving decisions.

3. Can We Use Angles to Make Robots Dance?

Did you know you can make a robot dance using angles? Just like humans move differently when they dance, robots can spin, turn, and move in different patterns based on the angles we program. Let's explore how you can create fun dance moves for your robot using angles and coding!

**4. What Happens When We Change the Angle of a Robot's Arm?**

Like those used in factories, robots with arms must lift, rotate, and place objects precisely. But how does changing the angle of a robot's arm affect its movements? Let's explore how different angles can change how a robot grabs and places things and how this is important for ensuring everything works smoothly.